

Simulating ecologically relevant hydrological indicators in a temporary river system



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ABSTRACT

Hydrological indicators (HIs) are commonly used in eco-hydrological studies (i.e. environmental flow and hydrological status assessment). Their computation is based on streamflow data, and if measured data are not available, hydrological models can be used to generate flow data. The present paper describes a study that aimed to predict streamflow in a temporary river and to analyze the general reliability of some hydrological indicators evaluated by using simulated data instead of measured flow data. The SWAT model was used to predict daily streamflow in a river section of the Celone river (Puglia, Italy). Several HIs characterizing the patterns of river flow or specific hydrological components were evaluated using observed and simulated streamflow. The results show that the SWAT model is able to simulate streamflow in temporary river systems, but its performance under extreme low flow conditions may be a weak point. When simulated streamflow time series were used, the replicability of the HIs evaluated using a rigorous statistical methodology ranged from good to limited. Good performance was found for the magnitude of discharge in wet months (average monthly flow from November to May), for the high flow indicators (annual maxima, 1-, 3-, 7-, 30-, 90-day mean flow) and timing, while limited performance was detected for low flow indicators (annual minimum 1-, 3-, 7-, 30-, 90-day mean flow) and the number of zero flow days. Better performance for low flow indicators was found after introducing the zero-flow threshold. This type of eco-hydrological study may contribute to characterizing the flow regime and its alterations in regions with scarce data.

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1. Introduction

Intensive agriculture has altered the quantity and the quality of waters in many areas around the world. Several studies have identified and described the interactions between agriculture and aquatic systems, agriculture and the atmosphere, and agriculture and the soil (Gordon et al., 2008). As the global demand of water for agriculture is increasing, it is necessary for ecologists and hydrologists to develop new ways of analyzing and managing changes in river regimes and in groundwater status.

Flow regime and its components, from extreme low flows to floods, exert a considerable influence on all the other environmental factors such as water chemistry, physical habitat, and biota. Due to its important role in a river ecosystem, ecologists define the flow regime as a “master variable” in freshwater systems (World Bank, 2008).

Water resources have to satisfy increasing demand for human needs, but dams, diversions, water abstractions may modify the natural flow regime of a river with consequences on ecological components and on water quality (Lake, 2007; Zoppini et al., 2010; Arthington, 2012). It is obvious that the complete natural condition of streamflow is not maintained. Therefore, it is necessary to answer to the following question: How much water does a river need? (Richter et al., 1997). In particular, water resource managers have to preserve the natural status of water, as well as its use. During recent decades, hydrologists and ecologists have been working on developing tools to quantify alterations to the hydrological regime and to answer the abovementioned question (Poff and Ward, 1990; Arthington and Pusey, 1993; Richter et al., 1996; Black et al., 2005; Poff et al., 2010). The classical concept of “environmental flows”¹; (EF) has been changed in recent years, from the simple definition of a “minimum” flow level to the consideration of

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¹ Environmental flows describe the quantity, timing and quality of water flow require to sustain freshwater and estuarine ecosystems and human livelihoods and well-being that depend upon these ecosystems (Brisbane Declaration, 2007).

all of the components describing the natural flow regime, including floods and droughts. A large number of environmental flow assessment methods have been developed (Tharme, 2003; Acreman and Dunbar 2004; Arthington et al., 2006, 2010); however, few cases have been implemented, especially within Mediterranean watersheds with intermittent river systems.

In this context, several hydrological indicators (HIs) have been defined to characterize the river flow pattern as well as specific hydrological components, which may lead directly or indirectly to an influence on biological communities (e.g. annual minimum and maximum 1, 3, 7, 30 and 90-day streamflows, zero-flow days, etc.) (Poff, 1996). These HIs are commonly used to evaluate hydrological alterations, comparing indicators before and after a river has been altered by human activities (Richter et al., 1996; Poff and Zimmerman, 2010; De Girolamo et al., 2015). In addition, HIs can be also used when environmental flow requirements have to be set (Poff et al., 2010).

All the indicators are evaluated using a long time series of measured streamflow on a daily or monthly basis. Daily data recorded over at least 20 years are particularly important for small Mediterranean catchments, where rainfall events tend to be unevenly distributed in time and space, leading to rapid flow regime changes. In the past, this kind of river, in which flow is often intermittent, were considered irrelevant and thus not monitored in several countries in the Mediterranean region; hence, the availability of measured data is currently scarce, especially in pristine conditions (Oueslati et al., 2015).

Where flow record data are insufficient because there are no flow gauging stations or the available records are not long enough to cover both un-impacted and impacted conditions, flow data can be generated with diverse approaches (Black et al., 2005). When these types of procedure are attempted, care should be taken to ensure that generated data do not differ from existing series in relation to the time periods they span. Hydrological models (i.e. HEC-HMS, HSPF, and SWAT) are commonly used for simulating a hydrological regime over a long period of time (Singh and Woolhiser, 2002). Even if parsimonious models exist (Wagener et al., 2001), most hydrological models require a large amount of data and are parameter-intensive (De Girolamo and Lo Porto, 2012) and require calibration (Sivapalan, 2003). In fact, prediction in ungauged basins (PUB), without calibration, remains a difficult problem, which demands knowledge and understanding. In addition, model results are affected by uncertainty (Refsgaard et al., 2007). The latter includes *conceptual model uncertainty* (due to the simplification of processes or to processes unknown to the modeler and not included in the model), *input uncertainty* (due to errors in input data or to the extension of point data to large areas), and *parameter uncertainty* (the result of the inherent non-uniqueness of parameters in inverse modeling). Although uncertainty analysis has received increasing attention over the last three decades, a significant proportion of the community is still reluctant to embrace the estimation of uncertainty in hydrological modeling, as stated by Pappenberger and Beven (2006). They identified arguments against uncertainty analysis adduced by several modelers (i.e. uncertainty analysis is difficult to perform, it is very subjective, it is not physically based, etc.), even if none of them are, in the end, tenable. Moreover, they suggest that one reason why the application of uncertainty analysis is not a normal and expected part of modeling practice is that mature guidance on methods and applications does not exist. In addition, most hydrological models have been developed for perennial rivers and present several limits in simulating extreme low flow conditions in intermittent rivers (Kirkby et al., 2011). As a result, the hydrological indicators evaluated on a simulated streamflow basis could not be realistic. In this case, the hydrological alterations estimated on the latter indicators could also be wrong.

In this context, the present paper describes a study which aims to (1) predict streamflow in a temporary river (Celone, Apulia in southern Italy) oriented to support eco-hydrological studies such as environmental flow and hydrological alteration assessment; and (2) analyze the general reliability of some hydrological indicators evaluated by using simulated data instead of measured streamflow data.

It should be highlighted that this study is not intended to be an evaluation of the capabilities of the SWAT model and the processes it predicts, nor a description of basin characteristics. Going beyond a typical hydrological simulation of a basin with a temporary river system, this research focused on the general capabilities as well as parameter uncertainties surrounding the replications of some hydrological indicators.

2. Study area

The study area is the Celone catchment, located in the Apulia region in southern Italy (Fig. 1). The drainage area is 317 km² and the main river course is 93 km long. The main economic activity in the area is agriculture, which is intensive only in the plains. The main farming systems are cereals (70%), mostly durum wheat, tomatoes (3%), vegetable (6%), olives (4%), and grapes (3%). Most of the production in the plain area (Tavoliere delle Puglie) is based on irrigation, although in the Apulia region water availability is very limited and depends on water inflow from neighboring regions (Campania and Basilicata). The upper part of the basin (Monti Dauni) is characterized by an irregular morphology, where forest lands (8%) and pasture (2%) are frequent and rainfed agriculture is practiced in an extensive way. Urban areas and set-aside land occupy a limited area (4%). The basin is characterized by a mean elevation of about 300 m, ranging from 0 to 1100 m. From 1990 to 2009, the average annual precipitation recorded in the basin was 625 mm, ranging from 840 mm in the mountainous area to 465 mm in the plain area. Rainfall is mostly concentrated in autumn and winter (from November to May); generally, it is unevenly distributed throughout the basin and during the dry season, from June to September, it is concentrated in a few events of short duration and high intensity and are extremely localized. The soil texture varies from sandy-clay-loam (36%), sandy-clay (1%) to clay (63%). The stream is incised in the upper basin, then on an alluvial plain downstream of the steeper reaches assumes a braided form; after that, the river resumes a sinuous course. The river system shows an intermittent character, with a pattern of extreme low flow or zero flow. During the dry months, surface flow is very low or absent and some river segments are completely dry or characterized by small isolated pools that succeed one another along the stream channel. From June to September, flash flood events are quite common; these floods have a very rapid rising stage and a short lag time (time between peak rainfall and peak discharge). From 1965 to 1995, the average streamflow recorded at Gauge 1 (Celone S. Vincenzo, 41°25'25"N; 15°24'31"E) (Q_m) was 0.48 m³s⁻¹. Currently, no streamflow data are available either from this gauging station or in other gauges in the study area.

In the watershed, major hydrological pressures from different levels come mainly from point source discharges provided from waste water treatment plants (WWTPs), water abstractions from the river and groundwater for agricultural water use, and a reservoir. The Capaccio reservoir (41° 25' 30" N; 15° 25' 25" E) was built for irrigation purposes in 2000; the water volume in 2015 was 16.8 Mm³ (May 2016) and its capacity at full supply is 25.82 Mm³. The Celone stream is the main inflow into the reservoir. Official data which quantify water abstraction from the river and water withdrawal from the groundwater are not available. However, irrigation practices are heavily used within this watershed. Volumes of waste

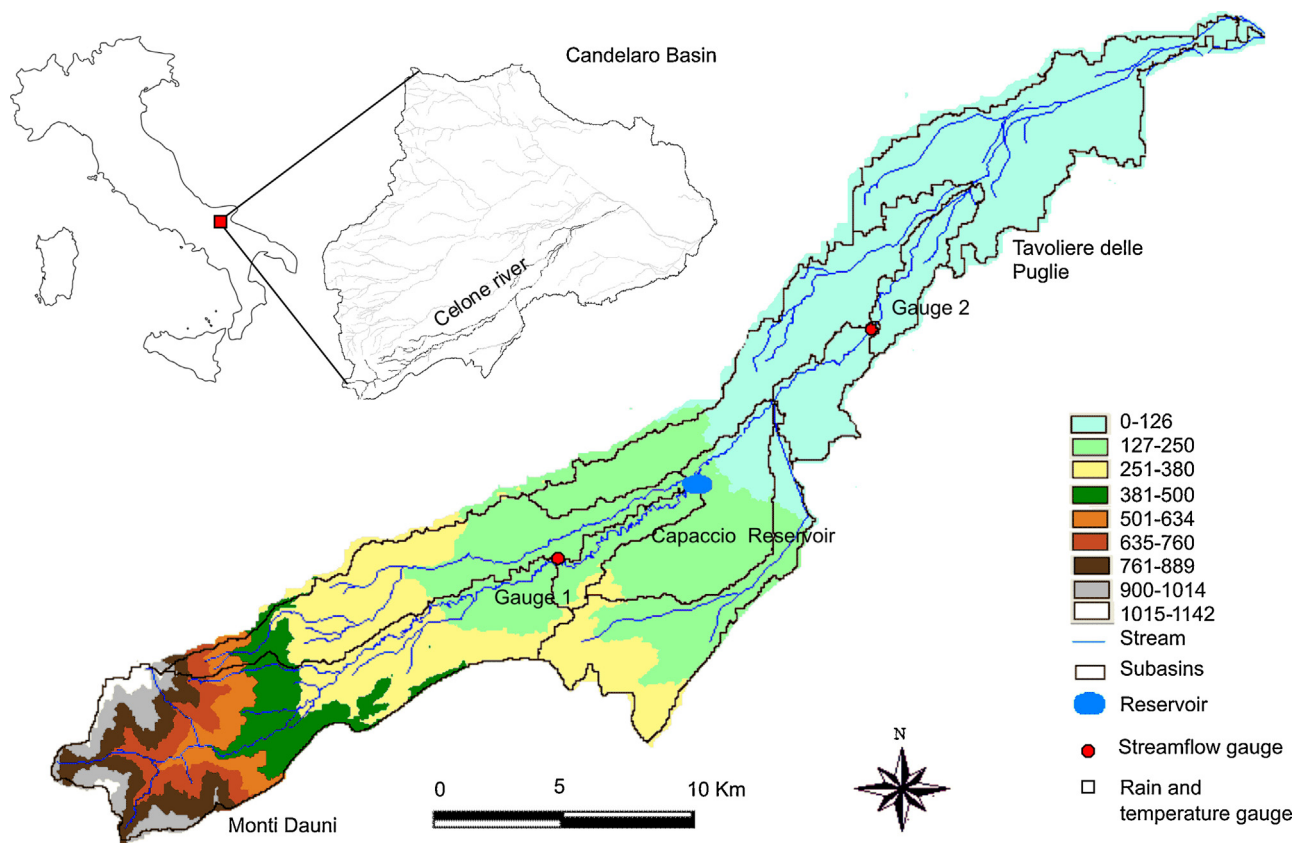


Fig. 1. Study area: Celone river basin. “Tavoliere delle Puglie” represents the plains and “Monti Dauni” the Apennines. Gauge 1 is also called “Celone S. Vincenzo” and Gauge 2 is called “Celone Ponte Foggia”.

water discharged by WWTPs are provided on a monthly basis from year to year. Significant changes in land use have not been recorded in recent decades.

In Italy, there is currently no standard methodology for assessing environmental flow (EF). Italian Legislative Decree no. 152/2006 states that the rules for EF calculation have to be defined by the individual region in accordance with the objectives defined by the River Basin Authority in the River Basin Plans. Furthermore, the Apulia Region has not imposed specific recommendations for water release from downstream dams.

Fig. 1. Study area: Celone river basin. “Tavoliere delle Puglie” represents the plains and “Monti Dauni” the Apennines. Gauge 1 is also called “Celone S. Vincenzo” and Gauge 2 is called “Celone Ponte Foggia”.

3. Material and methods

The hydrological processes in the study area were simulated using the Soil and Water Assessment Tool model (SWAT) (Arnold et al., 1998; Srinivasan et al., 1998) (in Fig. 1). The model was calibrated and validated using measured data over the period 1990–1996 at Gauge 1 (Celone S. Vincenzo 41°25′25″N; 15°24′31″E) (De Girolamo et al., 2015) and at Gauge 2 downstream (Celone Ponte Foggia 41°31′34″N; 15°32′42″E). Next, a large number of hydrological indicators (HIs) representative of the flow regime were calculated using simulated (Q_s) and measured daily flow data (Q_m) in a river section (Gauge 1 in Fig. 1). The corresponding indicators, computed with both Q_m and Q_s , were compared in order to analyze the general reliability of some hydrological indicators evaluated using simulated data instead of measured streamflow data. The common period of the two datasets covers

only a few years (from 1990 to 1996) because measured streamflow data were available at a daily time scale from 1965 to 1996, while daily climatic data required by the SWAT model were available from 1990 to 2010.

3.1. Modeling approach to simulate streamflow

To simulate hydrological processes, we used the SWAT model, version SWAT2005 with the Arcgis interface (Winchell et al., 2007). The model is able to simulate streamflow and water quality in agricultural, urban and pristine basins for historical conditions, future conditions (Abouabdillah et al., 2010; Krisanova and Srinivasan, 2015), and natural conditions (Kiesel et al., 2010). It is widely used for analyzing different scenarios (Ullrich and Volk, 2009). The model was applied with good results in several Mediterranean basins where flow data are scarce (De Girolamo and Lo Porto, 2012; Abouabdillah et al., 2014). The model also includes procedures for enabling sensitivity analysis and uncertainty analysis (van Griensven et al., 2002; Abbaspour et al., 2007).

The inputs used in the model and their relative sources are summarized in Table 1. In order to establish reaches with intermittent flows and to estimate anthropogenic impacts (i.e. water abstractions) in the study area, a survey campaign and several interviews with local citizens and farmers were done.

We ran the model on a daily time-step from January 1990 to December 2009, based on the availability of daily rainfall and temperature data. In this time period, only a few years of measured flow data were available. Before the dam was built (prior to 1996), streamflow was recorded on a daily time scale with several gaps, while from 2000 to 2010 (after dam construction), only monthly data on the volume released downstream by the dam were available, with several gaps.

Table 1
Model input data and their sources.

Variable	Origin	Scale	Information
Precipitation	Civil Protection Service Apulia Reg. Agency	Daily value (on basin scale)	3 rainfall stations (1990–2009)
Temperature	Civil Protection Service Apulia Reg. Agency	Daily value (on basin scale)	2 temperature stations (1990–2009)
Land use map	Corine Land Cover 2000 EU Project	ArcInfo format (scale 1:100000)	Minimum area digitalized 25 ha
Soil map	ACLA 2—FEOGA EU Project	ArcInfo format (scale 1:100000)	9 soil profiles
Management Practices	Consorzio per la Bonifica della Capitanata		Irrigation amount, tillage oper., fertilizers appl.
Digital Elev. Model	Apulia River Basin Authority	Arc Info grid format (40 × 40 m)	
Waste Water Treatment Plants discharges	Ecological Police (FG)	Average daily values	Daily discharges (m ³ /s)

In the plains (Tavoliere delle Puglie), an irrigation board (*Consorzio per la Bonifica della Capitanata*) operates with a well-defined irrigation system, providing water at a competitive cost. The Consortium is a non-profit organization and farmers have to pay only a contribution which is proportioned both to the surface served and on the volume of water taken from the hydrant (all the hydrants are equipped with water meters). In the areas served by the Consortium, we fixed the number of irrigation events and water amount on the farmers' interviews basis as input in the SWAT simulations.

The study area was divided into nine sub-basins. We selected the *Hargreaves and Samani* (1985) method for evaluating evapotranspiration because daily wind speed and daily relative humidity were not available. To calculate surface runoff, the Soil Conservation Service Curve Number method (SCS-CN) (*USDA-Soil Conservation Service*, 1972) was selected. A sensitivity analysis (SA) was performed in order to identify the most sensitive parameters with the LH-OAT method, which combines OAT (one factor at a time) and Latin Hypercube sampling (*van Griensven et al.*, 2002) for 27 hydrological parameters using the module incorporated in the version SWAT2005. For this analysis, we used the observed flow data at Gauge 1 (1990–1992). Some of the most sensitive parameters were: the initial SCS curve number for moisture condition II (CN2), the depth from the soil surface to the bottom of the layer (SOL_Z [mm]), the threshold depth of water in the shallow aquifer required for return flow to occur (GWQMN [mm]), the available water capacity of the soil layer (SOL_AWC [mmH₂O/mmsoil]), the soil evaporation compensation factor (ESCO), and the surface runoff lag time (SURLAG [days]).

From June to December, the river shows extreme low flow and it is not rare to observe a continuum where perennial reaches with low flow (a few liters per second), pools sustained by sub-surface flow, and completely dry reaches longitudinally coexist. Based on the results of a detailed study published by *Ippolito et al.* (1958) who analyzed the hydrogeological aspects of the study area and pointed out the bidirectional surface aquifer exchanges, the effective hydraulic conductivity in the main channel alluvium (CH_K) was fixed at 150 mmhr⁻¹.

The calibration and the uncertainty analysis were carried out at Gauge 1 by applying the Sequential Uncertainty Fitting version 2 (SUFI 2) procedure developed by *Abbaspour* (2011) included in the SWAT-CUP software (<http://www.neprastechnology.ca/>). SUFI 2 combines calibration and uncertainty analysis to find parameter uncertainties (*Abbaspour et al.*, 2007). A manual calibration was performed for Gauge 2.

The model was calibrated over the period 1990–1992 at Gauge 1 (Celone S. Vincenzo) and from 1990 to 1991 at Gauge 2 downstream (Celone Ponte Foggia). The model was validated from 1995 to 1996 at Gauge 1 and in 1994 at Gauge 2. Some results were presented in a previous work (*De Girolamo et al.*, 2015). The periods for calibration and validation were selected among those available by taking into account two questions: (i) how reliable are the input

data (rainfall versus streamflow)? and (ii) how does this affect the model? Our final aim in model calibration was not to maximize the statistical analysis of the selected performance criteria but to understand how the basin works and to reproduce the hydrological process dynamics adequately through the selection of realistic and optimal parameter values. Hence, we did not consider those flow record periods characterized by gaps and by several discrepancies between rainfall and measured streamflow because this could promote a focus on trying to get a good model fit to the data instead of trying to understand what was actually happening in the basin. On the other hand, *Sorooshian and Gupta* (1983) suggested that it is the quality of data, rather than the quantity, which may be the more important characteristic for a given data set. However, in the Celone catchment, frequent rainfall events are extremely localized in very small areas and, consequently, the hydrological regime response is limited or absent.

3.2. Definition of hydrological metrics

A general approach for describing hydrological regimes uses hydrological indicators based on the statistics of streamflow data (*Arthington*, 2012). The indicators describe specific characteristics of streamflow such as: (1) magnitude of discharge (amount of water moving past a fixed location per time unit), (2) duration (period of time associated with a specific flow condition), (3) timing (the regularity with which flows of a defined magnitude occur), (4) frequency (refers to how often a flow above a given magnitude recurs over some specified time interval), and (5) rate of change (refers to how quickly flow changes from one magnitude to another) (*Poff et al.*, 1997).

In the present study, 32 hydrological indicators (HIs) described by *Richter et al.* (1997) were used. They were calculated using Indicators of Hydrological Alterations software (IHA), version 7.1 (*The Nature Conservancy*, 2009).

The indicators were evaluated using both measured and simulated daily streamflow data. Table 2 shows all the indicators analyzed with a short description.

Table 2 List and description of the examined hydrological indicators (HIs). The HIs are grouped into five groups, describing ecologically relevant components of the hydrologic regime, such as magnitude (group 1), duration (group 2), timing (group 3), frequency (group 4) and rate of change (group 5).

3.3. Comparison of predicted vs. observed indicators

In this study, to analyze differences between the two series of indicators (simulated and observed), a rigorous methodology for assessing the statistical equality between two sets of values was applied. In the case at hand, the nature of analyzed data suggests the use of the paired (or coupled) *t*-test.

The *t*-test is the best suited statistical test to carry out comparisons between two groups (*Rice*, 2006), but it needs to fulfil a

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Group	Code	Name	Significance
1	MAJ	January mean flow	Average January flow
	MAF	February mean flow	Average February flow
	MAMar	March mean flow	Average March flow
	MAAP	April mean flow	Average April flow
	MAM	May mean flow	Average May flow
	MAJN	June mean flow	Average June flow
	MAJL	July mean flow	Average July flow
	MAAG	August mean flow	Average August flow
	MAS	September mean flow	Average September flow
	MAO	October mean flow	Average October flow
	MAN	November mean flow	Average November flow
	MAD	December mean flow	Average December flow
2	DL1	Annual minima, 1-day minimum	Magnitude of minimum annual flow of 1 day duration
	DL2	Annual minima, 3-day minimum	Magnitude of minimum annual flow of 3 day duration
	DL3	Annual minima, 7-day minimum	Magnitude of minimum annual flow of 7 day duration
	DL4	Annual minima, 30-day minimum	Magnitude of minimum annual flow of 30 day duration
	DL5	Annual minima, 90-day minimum	Magnitude of minimum annual flow of 90 day duration
	DH1	Annual maxima, 1-day mean	Magnitude of maximum annual flow of 1 day duration
	DH2	Annual maxima, 3-day mean	Magnitude of maximum annual flow of 3 day duration
	DH3	Annual maxima, 7-day mean	Magnitude of maximum annual flow of 7 day duration
	DH4	Annual maxima, 30-day mean	Magnitude of maximum annual flow of 30 day duration
	DH5	Annual maxima, 90-day mean	Magnitude of maximum annual flow of 90 day duration
	DL6	Number of zero days	Mean annual number of days having zero daily flow
	ML1	Base flow index	7-day minimum flow divided by mean flow for year
3	TH1	Date of maximum	Julian date of annual maximum flow
	TL1	Date of minimum	Julian date of annual minimum flow
4	DH6	High pulse duration	Mean or Median duration of high pulses
	FH1	High pulse count	Number of high pulses within each year
	FL1	Low pulse count	Number of low pulses within each year
5	RA1	Rise rate	Median of all positive differences between consecutive daily values
	RA2	Fall rate	Median of all negative differences between consecutive daily values
	RA3	Number of reversals	Number of times that flow switches from daily change positive (rising) to negative (falling).

number of assumptions to be reliably applied. The first assumption concerns the distributions of both the sets to be compared, which should be Gaussian; the second concerns the variances of the two sets, which should be equal (homoscedasticity). Strictly speaking, the *t*-test compares the average values of the two sets, but considering the aforementioned assumptions together with the outcome of the *t*-test, it is possible to check if the two compared sets can be considered statistically equal or not. Where the assumptions were not met, the Mann-Whitney test should be applied as a substitute for the *t*-test (Siegel, 1956). The Mann-Whitney test does not possess the same reliability characteristics of the *t*-test; this is why it is used only when the latter cannot be applied.

A large amount of statistical tests are available for checking the Gaussian (normal) distribution of a given dataset, each of them best suited according to some specific feature of the set to be analyzed (small/large dataset, the data type, etc.). For the case at hand, four different tests were applied, namely Shapiro-Wilk, Anderson-Darling, Lilliefors and Jarque-Bera (Razali and Wah, 2011; Jarque and Bera, 1981). The decision rule for accepting (or rejecting) normality is that the majority of methods accept the normality hypothesis (or rejected). After the assumptions check, the *t*-test (or the Mann-Whitney test) can be reliably applied to data and the statistical equality of the averages (hypothesis H0) of the two compared data sets can be assessed.

4. Results

4.1. Model performance

The performance of the model on daily time step run was evaluated using the Nash and Sutcliffe efficiency (NSE) (Nash and Sutcliffe, 1970), correlation coefficient (R^2), positive values of the percent bias (PBIAS) (Gupta et al., 1999), and RSR (RMSE-

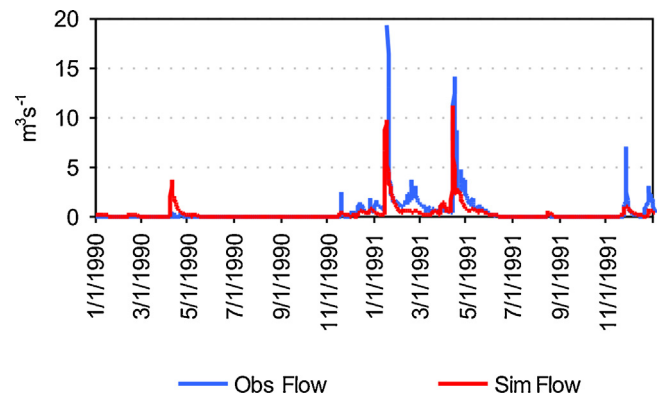


Fig. 2. Daily measured and simulated streamflow at Gauge 2. Calibration (1990–1991). NSE = 0.51; PBIAS = 24%; RSR = 0.70; R^2 = 0.83.

observations standard deviation ratio). For the calibration period, the NSE value at Gauge 1 was 0.74 and at Gauge 2 was 0.51; the R^2 values were 0.89 and 0.83; PBIAS was –25% and 24%; while RSR was 0.50 and 0.70, respectively. PBIAS indicates model underestimation bias, and negative values indicate model overestimation bias. Fig. 2 shows the simulated and observed flow for the calibration period at the Gauge 2. For the validation period (1995–1996), at Gauge 1 (Fig. 3) the NSE was 0.47, R^2 was 0.77, and RSR and PBIAS were 0.76 and 26%, respectively, while at Gauge 2 (1994) the NSE was 0.83, R^2 was 0.93, RSR and PBIAS were 0.41 and 10%, respectively. In their review of published literature, Moriasi et al. (2012) defined the model performance for streamflow simulated on a monthly time step “satisfactory” if $NSE > 0.50$ and $RSR \leq 0.70$ and, for measured data of typical uncertainty, if $PBIAS \pm 25\%$. These ratings should be adjusted to be more or less strict based on project-specific considerations, and appropriate relaxing and tightening of

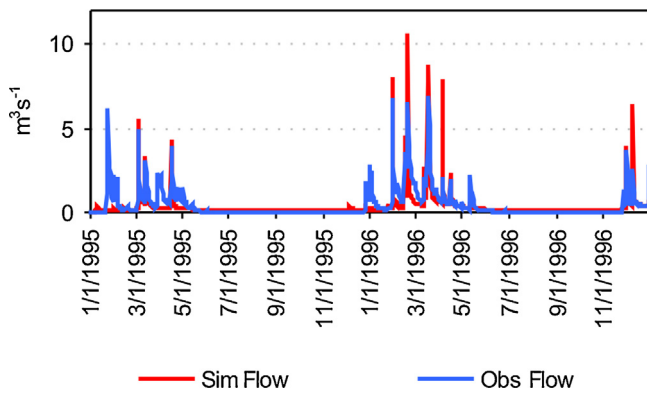


Fig. 3. Daily measured and simulated streamflow at Gauge 1. Validation (1995–1996). NSE = 0.47; PBIAS = 26%; RSR = 0.76; $R^2 = 0.77$.

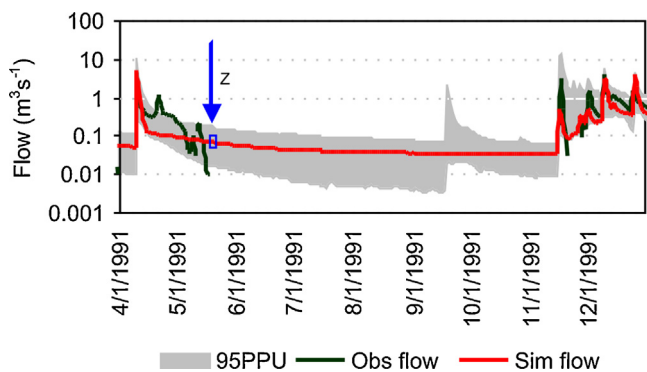


Fig. 4. Measured streamflow (Obs flow) and the “best simulation” (Sim flow). The grey interval is 95% prediction uncertainty (95PPU) for Gauge 1 (1990) (P-factor = 0.41 and R-factor = 0.20). The blue arrow shows the zero-flow threshold (Z). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

the standard should be performed for daily and annual time step evaluations, respectively (Engel et al., 2007; Arnold et al., 2012). In fact, the poorest results generally occurred for daily predictions, although this was not universal (Grizzetti et al., 2005). Based on these criteria, and considering the quality and quantity of our measured data, the simulation time step run (daily), and the project scope, we considered the results satisfactory.

The parameters, their range and the calibrated values are summarized in Table 3.

Fig. 4. Measured streamflow (Obs flow) and the “best simulation” (Sim flow). The grey interval is 95% prediction uncertainty (95PPU) for Gauge 1 (1990) (P-factor = 0.41 and R-factor = 0.20). The blue arrow shows the zero-flow threshold (Z).

The results of the uncertainty analysis, using the SUFI 2 procedure, are shown in Fig. 4, which represents the observed streamflow, the “best simulation” and the 95% prediction uncertainty (95PPU) for Gauge 1 (Celone S. Vincenzo). As the graph shows, the uncertainty interval is quite large. These results confirm the studies of Uhlenbrook et al. (1999), who reported that an uncertainty interval may be very large in extreme low flow conditions. The streamflow values of the calibrated simulation were used to calculate all the hydrological indicators.

4.2. Evaluation of hydrological indicators

The aforementioned statistical tests were used to analyze the differences between the two series of indicators, computed with measured data and simulated streamflow, respectively. Concerning the HIs describing the monthly magnitude of discharge (Group

1 in Table 2), it can be highlighted that the model fails in generating a prediction series resembling to the observed series in only four cases, namely for June mean flow (MAJN), July (MAJL), September (MAS), and October (MAO) (Table 4). For all the other indicators representing average monthly flow, the model was able to reproduce the observed data series. In fact, the distribution shapes, variances, and average values can be considered statistically all equal, stating the effectiveness of the applied model. It can be noticed that the four failed cases are those which showed a non-Gaussian distribution and heteroscedasticity (i. e. inequality of variances) when computed with the measured flow data, while the simulated data series always showed a Gaussian distribution.

If the same tests are applied to the indicators describing the frequency of low flow (Group 2 in Table 2: DL1, DL2, DL3, DL4, DL5), a clear departure can be seen between the statistics of the HIs based on observed data series and those of the model-predicted values (Table V).

This result is due to the marked differences between the minimum values of the observed streamflow data series (Q) and the minimum in the series of the predicted Q values (see Fig. 4, Fig. 5). On the contrary, all the indicators describing the frequency of high-flow indices (DH1, DH2, DH3, DH4 and DH5) show statistical equality with the corresponding metrics computed from the observed data series, demonstrating the effectiveness of the applied model in predicting the behavior of the physical process (Table 5). In conclusion, it should be noted that the model fails in correctly predicting the behavior of Q in correspondence with the driest months and consequently with the HIs related to periods of dryness. Therefore, accounting for this disagreement between observations and predictions, model outputs were adjusted by subtracting a suitable threshold from the predicted values. The threshold, called the “zero-flow” value throughout this paper, was fixed at 75 l s^{-1} (Fig. 4). It was determined by simultaneously comparing the measured and simulated streamflow data for the driest year recorded in the study period. This value is specific for river section 1 (Gauge 1 in Fig. 1); it depends on the local conditions such as river bed permeability and hydraulic conductivity, in addition to the intrinsic limits of the hydrological model used for the simulations or to an underestimation of the water abstractions for the river. In a first methodological approach, the threshold was subtracted from all daily values, while in the second approach, the threshold was subtracted from the simulated daily streamflow values only from June to October.

After such modifications, the resulting datasets were compared with the validation set (HIs computed with measured data). The overall behavior was not uniform; in fact, September and October mean flow (MAS, MAO) was statistically equal to the validation dataset. The observed June and July mean flow (MAJN, MAJL), were different from their corresponding simulated values at a tested significance level of 95%, while they were equal to their corresponding simulated values at a significance level of 99%. Therefore, an approximate similarity was observed between the compared datasets, at least for the average values. Only the results concerning the four dry months are reported in Table 6.

Regarding the statistics related to the other HIs linked to periods of dryness (MAAG, DL1, DL2, DL3, DL4, DL5), both the first and the second approach showed that the correction was effective at making the simulated datasets comparable with the observed original data, as demonstrated by the results of the applied tests.

Moreover, the hydrological indicators describing timing, frequency and rate of change computed with simulated streamflow were subjected to the abovementioned data transform to check, once more, the effectiveness of the applied methodology. Indeed, all of them, namely Date min (TL1), Date max (TH1), Hi pulse count (FH1), Rise rate (RA1), Fall rate (RA2), Number of reversals (RA3) and

Table 3
Swat model parameters, their range and calibrated values.

Parameter	Rank	Description	Calibrated values	Range
CN2	1	SCS runoff curve number	54–88 ^a	35–98
SOL_Z	2	Depth from soil surface to bottom of layer [mm]	150–500 ^a	0.001–3500
GWQMN	3	Threshold depth of water in the shallow aquifer required for return flow to occur [mm]	0–10 ^b	0–5000
CANMX	4	Maximum canopy storage [mm]	3–7 ^a	0–100
SOL_AWC	5	Available water capacity [mm H ₂ O/mm soil]	0.08–0.16 ^a	0.00–1.00
ESCO	6	Soil evaporation compensation factor	0.35	0–1
BLAI	7	Maximum potential leaf area index [m ² /m ²]	1.25–5	0.50–10
SURLAG	8	Surface runoff lag coefficient [days]	7	0–10
GW_REVAP	9	Groundwater “Revap” coefficient	0.08–0.16 ^b	0.02–0.20
ALPHA_BF	10	Baseflow alpha factor [days]	0.37–0.9 ^b	0.00–1.00

^a Value varies according to input data (soil, land use).

^b Value was adapted in subbasins depending on their location: Monti Dauni (upstream Gauge 1), Tavoliere delle Puglie (downstream Gauge 1).

Table 4
Statistical test outcomes for observed and predicted HIs describing the average monthly flow. Y is for a Gaussian distribution, equal variances, and equal averages; N is for a non-Gaussian distribution, different variances, and different averages.

Hydrological Indicator	Gaussianity (Y/N)	Homosched. (Y/N)	Average equality (Y/N)
MAJ_obs	Y	Y	Y
MAJ_pred	Y		
MAF_obs	Y	Y	Y
MAF_pred	Y		
MAMar_obs	Y	Y	Y
MAMar_pred	Y		
MAAP_obs	Y	Y	Y
MAAP_pred	Y		
MAM_obs	Y	Y	Y
MAM_pred	Y		
MAJN_obs	Y	N	N
MAJN_pred	Y		
MAJL_obs	N	N	N
MAJL_pred	Y		
MAAG_obs	N	Y	Y
MAAG_pred	Y		
MAS_obs	N	N	N
MAS_pred	Y		
MAO_obs	N	N	N
MAO_pred	Y		
MAN_obs	Y	Y	Y
MAN_pred	Y		
MAD_obs	Y	Y	Y
MAD_pred	Y		

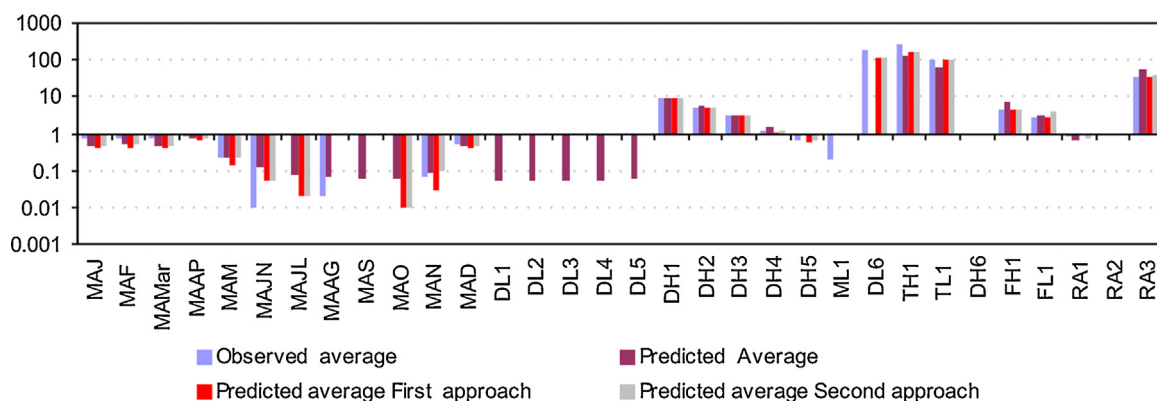


Fig. 5. Comparison between indicators computed with the observed flow data, SWAT simulated data (i), model simulated with the zero flow threshold first approach (ii); model simulated with the zero flow threshold second approach (iii).

Base flow (ML1) were statistically similar to the originally observed values, at least at the 1% of level of significance for both approaches.

Fig. 5 shows the hydrological indicators computed with the observed flow data, SWAT simulated data (i), model simulated with

the zero flow threshold first approach (ii); model simulated with the zero flow threshold second approach (iii).

Fig. 5. Comparison between indicators computed with the observed flow data, SWAT simulated data (i), model simulated with

Table 5

Statistical test outcomes for observed and predicted HIs describing the frequency of low flow and high flow. Y is for a Gaussian distribution, equal variances, and equal averages; N is for a non-Gaussian distribution, different variances, and different averages.

Indices	Gaussianity (Y/N)	Homoscedasticity (Y/N)	Average equality (Y/N)
DL1_obs	N	N	N
DL1_pred	Y		
DL2_obs	N	N	N
DL2_pred	Y		
DL3_obs	N	N	N
DL3_pred	Y		
DL4_obs	N	N	N
DL4_pred	Y		
DL5_obs	N	N	N
DL5_pred	Y		
DH1_obs	Y	Y	Y
DH1_pred	Y		
DH2_obs	Y	Y	Y
DH2_pred	Y		
DH3_obs	Y	Y	Y
DH3_pred	Y		
DH4_obs	Y	Y	Y
DH4_pred	Y		
DH5_obs	Y	Y	Y
DH5_pred	Y		

Table 6

Statistical test outcomes for observed and predicted HIs (modified by subtracting the zero flow threshold) describing the monthly mean flow in the dry months. Y is for equal averages and N is for different averages.

Indices	Average equality (Y/N)
MAJN_obs	N
MAJN_Modpred	
MAJL_obs	N
MAJL_Modpred	
MAS_obs	Y
MAS_Modpred	
MAO_obs	Y
MAO_Modpred	

the zero flow threshold first approach (ii); model simulated with the zero flow threshold second approach (iii).

5. Discussion

5.1. Modeling streamflow in intermittent rivers: lessons learned

The Celone stream, as with most temporary rivers, is characterized by high longitudinal variation in river flow (Larned et al., 2010; Gallart et al., 2012). Flow appears and disappears longitudinally depending on the substrate of the channel, and on the lithological and geological formations that influence the bidirectional surface-aquifer exchanges. Hence, flow regime characterization based on hydrological indicators is site-specific. This means that the hydrological indicators computed for one river section may vary upstream and downstream in response to the longitudinal changes in streamflow. It is well-known that the longitudinal variation of flow in intermittent streams is a peculiarity which makes the simulation of the streamflow more difficult here than in perennial rivers. In these cases, when calibrating the model, it is important to understand the hydrological response of the basin, so the model input parameters can be fixed within a realistic uncertainty range. Automatic procedures for calibrating the model cannot substitute actual physical knowledge of the watershed (Arnold et al., 2012).

Although the simulation of low flow is needed to improve water management in order to take measures to mitigate the socioeconomic and ecological impacts of water shortages (Nicolle et al., 2015), low flow has received only limited attention in the lit-

erature compared, for instance, to flood forecasting (Cloke and Pappenberger, 2009). Low flow simulations with hydrological models remain a difficult task, since processes conditioning low flows may depend on several factors such as physical basin characteristics, the relationship between surface water and groundwater, model structure, and management practices. A number of techniques (i.e. multi-model approaches) have been developed to be used in conjunction with a hydrological model to improve simulations and decrease modeling uncertainty (Georgakakos et al., 2004; Velázquez et al., 2011). Several attempts have also been made to improve the low flow simulation performance of the SWAT model. Lévesque et al. (2008) used a seasonal calibration scheme, in which summer and winter data were used to calibrate the model separately in two seasons. They obtained improved performance in the dry season when using summer observations; however, when they used the wet season data, no advantage was achieved compared with the traditional calibration method based on all available data. White et al. (2009) allowed SWAT to use a different curve number (CN) in growing and dormant seasons, obtaining a slightly improved daily model performance (NSE increased from 0.42 to 0.44). Muleta (2012) also adopted the seasonal calibration method to calibrate the SWAT model, but obtained small NSE values in general and, especially for the validation period, NSE was remarkably smaller than that of the calibration period, which might have been caused by over-fitting. Zanghi et al. (2015) developed a modified SWAT model (SWAT-SC) incorporating seasonal calibration in a subtropical monsoon climate area with a perennial river. For a lowland catchment, Guse et al. (2013) identified the groundwater and evaporation parameters as the main reasons for the poor performance of low flow reproduction by SWAT. They proposed a modification to the SWAT groundwater module to improve the representation of low flow.

In the present work, we simulated daily streamflow and all the components of the hydrological regime using hydrological indicators. The results show the satisfactory efficiency of the model in the daily time step run, while the replicability of the HIs ranged from good to limited. Good performance was found for wet months and peak flows, while limited performance was noted for low flows. Different factors responsible for these discrepancies were identified: errors in inputs (meteorological data, water abstractions, waste water discharges), spatial variability of data, errors in observed discharge data, model structure, and uncertainties. All these factors

affect the model's ability to replicate observed flows, especially low flow (Refsgaard et al., 2007; Pfannerstill et al., 2014). In addition, in the Celone river basin, only staff gauges were used to detect measures of water level, and the measurements of extreme low flow could be unreliable if the flow was divided in rivulets across the section.

The lack of data is a general problem in catchment modeling within the Mediterranean region (Oueslati et al., 2015). Even if several studies and EU research programs have highlighted the importance of these river systems in southern Europe (Nikolaïdis et al., 2013; Prat et al., 2014), the monitoring programs adopted by the Italian Regional Authorities are not yet sufficient to compose a complete database for eco-hydrological studies. Daily climatic data constitutes one of the most important datasets to simulate hydrological processes. The extension of the time series as well as gauging station locations has to be accurately analyzed before starting with model set-up. In this study area, the climatic gauging stations were few and not spatially well-distributed across the watershed. In addition, the time series contain several gaps. The Celone river basin shows a very heterogeneous hydrological response over time and space, resulting in limitations in streamflow modeling and large uncertainties in predictions, especially during the dry months. In this period, it is quite frequent that convective clouds concentrate rainfalls mainly in small areas in the mountains. Hence, as only a few rainfall gauging stations cover the whole basin, it may happen that rain recorded in a location is attributed in the modeling to an area much larger than that actually affected by the event, making it difficult to calibrate the model.

The “no-flow” condition is a crucial point in the hydrological characterization of a flow regime as well as in the calculation of hydrological indicators. A discrepancy of a few liters per second between the measured and simulated streamflow may be negligible in perennial streams, but may assume great relevance in intermittent river systems. Furthermore, the hydrologic indicators which were evaluated on the basis of overestimated flow could lead to incorrect river classification (De Girolamo et al., 2015) and hence to misleading results in eco-hydrological studies. Several researchers have pointed out the importance and the role of the flow type in macro-invertebrate community assemblages (Munné and Prat, 2011; Bonada et al., 2007; Kändler and Seidler, 2013). Buffagni et al. (2009) noted that an increase in lentic flow-related habitats may determine a decrease in the values of metrics used to evaluate ecological status (ES). Therefore, if these conditions are due to the natural variability in streamflow, a correction of ES assessment systems is needed in order to avoid an underestimation of ecological quality. As the discrepancy between measured and simulated flow could be attributed to several factors, as described above, expert knowledge is required to overcome the limits in simulating the hydrology of intermittent rivers.

In this work, the zero-flow threshold was introduced and applied for the summer months. The zero-flow threshold is the simulated streamflow that corresponds to a measured streamflow equal to zero in a river section. The streamflow values of the best SWAT simulation were modified for the dryer months, subtracting the zero-flow threshold from the daily values. As a result, the statistical performance of all hydrological indicators was improved. The results show that model prediction does not need a correction for months with high and moderate wet regimes, while it appears to be needed for dry months.

The River Basin Authority has not yet established a protocol for flow release. As a consequence, dam operators do not maintain adequate water flow to sustain the river ecosystem downstream of the reservoir. Hence, the present work contributes to streamflow management by simulating both natural/impacted streamflow and hydrological alterations attributable to human influence, which are

fundamental to estimate the environmental flow downstream of the reservoir.

6. Conclusions

Although temporary streams are generally not regulated by large dams, many anthropogenic activities may impact the flow regime, and these alterations may be important for river ecology. While dam operations affect all the components of the streamflow regime (magnitude, timing, duration, etc.), diversions or direct water abstraction both for irrigation, livestock, and domestic water supply mainly affect the low flow components. In particular, the timing, frequency, and duration of zero flow or extreme low flow may be strongly altered.

Hydrologic indicators commonly used to characterize the main features of the regime are easy to compute where discharge data from gauging stations are available. In situations with a scarcity, absence, or paucity of data, modeling techniques are generally employed, but these are conditioned by confidence in the modeling performance in order to effectively use environmental models for management and decision-making purposes. In particular, the assumptions may prove to be weak; therefore, expert evaluation of the results is necessary.

In the present work, the SWAT model was used to predict streamflow in the Celone watershed. Although it requires a large set of environmental data and is time-consuming, it was shown to be a useful tool in eco-hydrological studies and catchment management. This is because the model is able to simulate both natural and impacted streamflow, which are fundamental for estimating hydrological alterations and environmental flow downstream of the reservoir. However, extreme low flow conditions may be a weak point in the simulations.

The results of this work show that the replicability of HIs ranges from good to limited. Good performance was found for wet monthly mean flow and high flow conditions, while limited performance was detected for low flow and a number of zero flow days. A number of factors that cause such discrepancies were identified, such as errors in meteorological inputs and errors in observed discharge data. Additionally, model structure, spatial variability of the data, and uncertainties, which affect the model's ability to replicate observed flows (especially with extreme low flow), also contribute to the discrepancies. However, these disagreements do not necessarily indicate a shortcoming in this hydrological model for general hydrologic simulations, but it does point to limitations in how such models can be effectively applied in eco-hydrological studies. It should be stated that the model performance for temporary rivers could be improved by using a “zero flow” threshold. Consequentially, this could potentially lead to better performance for low flow indicators. This type of integrated eco-hydrological study contributes to improving knowledge of hydrological process simulations in catchments with temporary river systems and in catchment management with a specific focus on environmental flow requirements in the Celone river basin.

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Appendix A. Supplementary data

Supplementary data associated with this article can be found, in the online version, at <http://dx.doi.org/10.1016/j.agwat.2016.05.034>.

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